

Boing Blockchain — Implementation Roadmap

 **Everyday users:** testnet is live. Mainnet is still ahead. Track phases here; day-to-day use is [wallet](#) + [explorer](#).

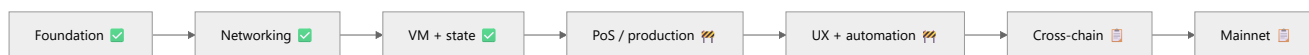
 **Developers:** Foundation and networking crates are in-tree. Remaining work is listed by phase — do not treat TBD vision docs as shipped.

 **Operators:** public RPC + Fly cluster are the current hosted path ([FLY-IO.md](#)).

Priorities: Security → Scalability → Decentralization → Authenticity → Transparency → True QA

Goal: Build an authentic, efficient L1 with unique innovations.

Aligned with: [BOING-BLOCKCHAIN-DESIGN-PLAN.md](#), [DEVELOPMENT-AND-ENHANCEMENTS.md](#)



Quick Start

```
# Build the workspace
cargo build

# Run the node
cargo run -p boing-node

# Run tests
cargo test
```

Project Structure

```

boing-network/
├── Cargo.toml           # Workspace root
├── crates/
│   ├── boing-primitives/ # Types, hash, crypto, signatures
│   ├── boing-consensus/  # PoS + HotStuff BFT
│   ├── boing-state/      # Verkle tree, state store
│   ├── boing-execution/ # VM, parallel scheduler
│   ├── boing-tokenomics/ # Block emission, fee split, dApp incentives
│   ├── boing-governance/ # Phased governance (time locks)
│   ├── boing-automation/ # Scheduler, triggers, executor incentives
│   ├── boing-qa/         # Protocol QA (Allow/Reject/Unsure)
│   ├── boing-cli/        # boing init, dev, deploy
│   ├── boing-p2p/        # libp2p networking
│   └── boing-node/       # Node binary
├── docs/                # All project documentation
├── fuzz/                # Cargo-fuzz harness (boing-primitives)
└── website/             # boing.network site (Astro + Cloudflare)

```

Phase 1: Foundation (Weeks 1–8)

1.1 Primitives — COMPLETE ✓

- ✓ Hash, Hasher (BLAKE3)
- ✓ AccountId, Account, AccountState
- ✓ Transaction, TransactionPayload, AccessList
- ✓ Block, BlockHeader
- ✓ Ed25519 signature; SignedTransaction; sign/verify
- ✓ Unit tests (hash, tx id, signatures, access-list conflicts)

1.2 Consensus — COMPLETE ✓

- ✓ HotStuff state machine (Propose → Vote → Commit)
- ✓ Quorum ($2f+1$); single-validator and multi-validator modes
- ✓ Unit tests (propose_and_commit, propose + vote flow)
- ✓ Leader rotation (round-robin)
- ✓ Simulate 4 nodes, 1 Byzantine

1.3 State Store

- ☒ In-memory `StateStore` (HashMap)
- ☒ Sparse Merkle tree for state root
- ☒ Transaction root (Merkle tree of tx IDs via `tx_root`)
- ☒ State root from Sparse Merkle
- ☒ Apply/revert batch (checkpoint/revert)

1.4 Execution (Minimal)

- ☒ VM: `Transfer` execution
- ☒ `TransactionScheduler` (access-list batching)
- ☒ Unit tests (scheduler batches)
- ☒ `BlockExecutor` (execute block of txs)
- ☒ Gas metering stub (fixed per tx type)
- ☒ Nonce validation

Phase 2: Networking & Integration (Weeks 9–16)

2.1 P2P Layer

- ☒ libp2p swarm setup (TCP, noise, yamux, TLS)
- ☒ Gossip: blocks, transactions (gossipsub)
- ☒ Peer discovery (mDNS)
- ☒ Request/response: get blocks by hash/height
- ☒ Block propagation and import (full node receives via gossip)

2.2 Advanced Decentralized Peer Discovery (see [DECENTRALIZATION-AND-NETWORKING.md](#))

- ☐ DHT-based discovery (Kademlia); minimize fixed bootnode reliance
- ☐ Gossip-first overlay: random peer selection, active probing
- ☐ Bootnode rotation (governance or community-funded)
- ☐ Peer scoring/reputation for Sybil/eclipse resistance
- ☐ WebRTC/WebSockets for browser light clients; decentralized signaling
- ☐ Incentivized relayers; DHT rendezvous for NAT traversal

2.2b Decentralized WebRTC Signaling (see [DECENTRALIZATION-AND-NETWORKING.md](#))

- ☐ Signaling smart contract (offer/answer exchange, event emission)
- ☐ Spam prevention: on-chain rate limiting, deposit/stake for offers, identity/reputation gate
- ☐ IPFS/Filecoin integration for large SDP payloads; on-chain CID pointers
- ☐ DHT announcement of WebRTC capability; recipient lookup
- ☐ Incentivized STUN/TURN servers; protocol rewards for relay providers
- ☐ STUN/TURN on-chain registry with staking; reputation score from uptime, latency, success rate
- ☐ STUN/TURN slashing for poor performance; SDK server selection by reputation

2.3 Node Integration

- ☒ Mempool (transaction pool)
- ☒ Wire consensus ↔ execution ↔ state
- ☒ Block production loop (BlockProducer + produce_block_if_ready)
- ☒ Chain state (ChainState, genesis, append)
- ☒ CLI: `--validator`, `--rpc-port`, `--data-dir`
- ☒ JSON-RPC HTTP server (boing_submitTransaction, boing_chainHeight)
- ☒ Validator loop (produce blocks every BLOCK_TIME when --validator)
- ☒ Block import and validation (validate_and_execute_block, import_block)

2.4 End-to-End

- ☒ Single node: produce blocks with txs
- ☒ Two nodes: sync blocks (P2P gossip + block import)
- ☒ Multi-node testnet (4+ validators)

Phase 3: Verkle & VM (Weeks 17–28)

3.1 Verkle Tree / Sparse Merkle

- ☒ Sparse Merkle tree (insert, get, delete, root)
- ☒ State root from Sparse Merkle
- ☒ Proof generation (MerkleProof, prove/verify)
- ☒ Stateless client verification (boing_getAccountProof, boing_verifyAccountProof RPC)

3.2 Custom VM

- ☒ Bytecode spec (PUSH, ADD, SUB, MUL, MLOAD, MSTORE, SLOAD, SSTORE, JUMP, JUMPI, RETURN, STOP)
- ☒ Interpreter (deterministic stack machine)
- ☒ Gas metering per opcode
- ☒ Contract deployment and call

3.3 Parallel Execution (Full)

- ☒ Execute Transfer-only batches in parallel (rayon)
- ☒ Conflict detection at runtime (sanity check)
- ☒ Benchmarks: sequential vs parallel (criterion)

Phase 4: PoS & Production Readiness (Weeks 29–40)

4.1 Proof of Stake

- ☒ Validator set from stake (`StateStore::top_stakers`)
- ☒ Staking transactions (bond, unbond)
- ☒ Slashing conditions (equivocation detection in consensus)
- ☒ Rewards distribution (block emission to proposer)
- ☒ Sustainable tokenomics (design + boing-tokenomics crate: emission curve, fee split, burn, dApp cap)

4.2 Core Innovations (from Design Plan)

- ☐ Native Account Abstraction (protocol-level)
- ☒ Adaptive gas model (GasConfig, multiplier with predictable caps)
- ☒ Phased governance (boing-governance: proposal → cooling → execution)
- ☒ Transparent slashing with appeal

4.3 Hardening

- ☒ Fuzz testing (cargo-fuzz; fuzz/ crate, bincode Block/Transaction)
- ☒ Property-based tests (proptest: balance preservation, parallel vs sequential equivalence)
- ☐ Security audit (external); continuous audit cadence post-mainnet

- ☒ Documentation, runbook

4.4 Randomness & Fair Ordering

- ☒ VDF/VRF for verifiable randomness (leader selection; RFC 9381 ECVRF + `leader_from_vrf` / stub path)
- ☒ Replace or augment round-robin with VDF-driven selection (`leader_from_vrf` stub)

4.5 Security Standards (see [SECURITY-STANDARDS.md](#))

- ☒ DDoS resistance: RPC rate-limiting (`RateLimitConfig`, governor crate)
- ☒ Disk persistence (chain + state via `--data-dir`)
- ☒ P2P connection cap per remote IP (`RateLimitConfig.connections_per_ip` , `--max-connections-per-ip` ; signed tx gossip + mempool admit on receive — see [RUNBOOK.md](#) §8.1)
- ☐ Formal verification pipeline for critical contracts (staking, DeFi, automation)
- ☐ Bug bounty program
- ☒ Incident response runbook ([RUNBOOK.md](#) §6)
- ☐ Post-quantum cryptography research and integration path
- ☒ HD wallet path type (`HdPath` in `boing-primitives`)

4.6 Protocol Quality Assurance (QA) — see [QUALITY-ASSURANCE-NETWORK.md](#)

- ☒ **boing-qa crate:** Rule types (Allow / Reject / Unsure), central rule registry, deterministic checks (bytecode size, opcode whitelist, well-formedness, purpose categories including meme/community/entertainment)
- ☒ **Known edge-case resolution:** Implement §11 mappings so automation handles meme leniency, ambiguous declaration, new/unknown pattern; blocklist support
- ☒ **Node integration:** Run QA at submit and/or mempool insert for `ContractDeploy`; structured rejection (`rule_id`, message); optional `boing_qaCheck` RPC
- ☒ **Execution defense in depth:** QA check in `execute_contract_deploy` before `set_contract_code`
- ☒ **Community QA pool:** Pending QA queue (`boing_qa::pool`), pool members, voting, max time T, default outcome at expiry; on-chain integration pending
- ☐ **Governance of rules:** Add/modify rules via time-locked governance; versioned rule sets
- ☒ **Purpose in payload:** `ContractDeployWithPurpose` variant; `check_contract_deploy_full` with `RuleRegistry`
- ☒ **Always-review categories:** Policy categories that force Unsure; configurable via `RuleRegistry`
- ☒ **Scam patterns:** Byte-sequence blocklist (legitimacy heuristic) in `RuleRegistry`
- ☒ **Soft rules:** Example: "other" + minimal description → Unsure

- ☒ **Additional enhancements (optional):** Pre-flight SDK/CLI (boing_qaCheck RPC + SDK `client.qaCheck()`), public QA metrics, appeal path, canonical malice definition, deployer checklist (in progress)

Phase 5: UX & Human-Centered Innovations (Weeks 41–52)

5.1 Frictionless UX

- ☒ Transaction simulation API (boing_simulateTransaction)
- ☒ Gasless transactions (paymaster / meta-tx types: PaymasterConfig, SponsoredTransaction)
- ☒ Human-readable signing (display_for_signing on SignedTransaction, SignedIntent)
- ☐ Transaction simulation API (pre-flight)
- ☐ Pre-confirmation / outcome guarantees where feasible
- ☐ Automatic chain routing (cross-chain UX)

5.1b Intent-Based Execution (see [DEVELOPMENT-AND-ENHANCEMENTS.md](#))

- ☒ Intent signing format (SignedIntent, IntentPool, boing_submitIntent RPC)
- ☐ Meta-router: orchestrate optimal cross-chain path
- ☐ Solver/executor integration for intent fulfillment

5.2 Trust & Verification

- ☒ Intent-based transaction format (SignedIntent, boing_submitIntent)
- ☐ Simulation service / SDK
- ☐ Contract/address attestation hooks
- ☐ Verified display for addresses and amounts

5.3 Recoverability

- ☒ Social recovery types (Guardian, RecoveryRequest in boing-primitives)
- ☐ Protocol-level recovery flows
- ☐ Optional MPC / threshold custody integration points

5.4 MEV & Cost Predictability

- ☐ VDF-based fair ordering (or equivalent)
- ☐ Encrypted mempool options

- ☐ Predictable gas model
- ☐ MEV-aware transaction design

5.5 Developer Experience

- ☒ Boing CLI (`boing init` , `boing dev` , `boing deploy`)
- ☒ RPC API specification ([RPC-API-SPEC.md](#))
- ☐ Local/devnet parity for testing
- ☐ Cross-chain simulation support

5.6 Success-Based dApp Incentives & Boing SDK

- ☒ dApp incentive formula (DappMetrics, `dapp_incentive`, `VALUE_CAP_PER_DAPP`)
- ☐ Per-dApp owner cap (governance parameter)
- ☐ Success metrics (tx count, fees, volume)
- ☐ Automated payout distribution
- ☐ Dynamic fee allocation & developer royalties (dApp-specified splits)
- ☐ Reputation-based resource access (priority processing, discounted gas)
- ☐ Boing SDK (`boing init` , `boing dev` , `boing deploy`)
- ☒ Metrics registration API

5.6b Boing Studio & AI-Assisted SDK

- ☐ Boing Studio IDE (Remix-style: templates, debugging, one-click deploy)
- ☐ AI-assisted SDK (code generation, vulnerability scanning, optimization)

5.7 Decentralized Automation

- ☒ Native scheduler (CronSchedule, Scheduler)
- ☒ Execution verification (ExecutorAttestation, ZkpProof, FraudProof, OracleAttestation)
- ☒ Trigger-based execution hooks
- ☒ Decentralized executor incentives (`executor_reward`, `executor_slash` in `boing-automation`)
- ☐ User-facing automation SDK

Phase 6: Technical Innovations for Ecosystem (Weeks 53–64+)

6.1 Cross-Chain Primitives

- ☐ Light client verification at protocol level
- ☐ Bridge standards (IBC-style or custom)
- ☐ Cross-chain messaging primitives
- ☐ Trust-minimized bridge design (ZKP/MPC relayers with slashing)
- ☐ Decentralized oracle networks for external data

6.2 Stateless & Light Clients

- ☒ Verkle proof generation and verification (MerkleProof)
- ☐ Light client sync protocol
- ☐ Mobile-friendly validation
- ☐ Compact proof specs (for ecosystem adoption)
- ☐ Browser-based Wasm light clients

6.3 Intent & Advanced Execution

- ☒ Intent signing format (SignedIntent, IntentPool, boing_submitIntent)
- ☐ Solver/executor integration points
- ☐ Pre-confirmation API
- ☐ Commit-reveal for MEV-sensitive ops

6.4 Compliance & Real-World

- ☐ Privacy-preserving attestation design
- ☐ Off-ramp / fiat integration points
- ☐ Payment rail primitives
- ☐ Compliance-friendly APIs (selective disclosure)

6.5 Open Standards

- ☒ RPC/API spec ([RPC-API-SPEC.md](#))
- ☐ Bridge protocol documentation
- ☐ Reference implementations for interoperability

6.6 Decentralized Storage & Network Monitoring

- ☐ Filecoin/Arweave integration for permanent archival (chain state, tx logs)
- ☐ SDK tools for IPFS/Filecoin dApp frontend deployment
- ☐ Network topology monitoring (node distribution, decentralization dashboards)

Phase 7: Testnet & Mainnet (Ongoing from Phase 4)

7.1 Public Testnet

- ☐ Public testnet launch
- ☐ Testnet incentive program (sustainability-first): use [TESTNET.md](#) Part 3 for readiness checklist, incentive design (validators, developers, users), and 2–4 week duration.
- ☐ Validator onboarding
- ☐ boing.finance integration (testnet)
- ☐ Community engagement: grant programs, hackathons, educational content

7.2 Mainnet

- ☐ Token distribution (fair launch)
- ☐ Bridge to boing.finance and other chains
- ☐ Mainnet launch
- ☐ Governance activation

Innovation Summary (from Design Plan)

Category	Key Items
UX	Native AA, gasless, human-readable signing, simulation, social recovery
Trust	Intent display, attestation, verified addresses
MEV	VDF ordering, encrypted mempool, predictable gas
Technical	Verkle stateless clients, light-client bridges, intent architecture
Ecosystem	Open standards, bridge protocols, reusable specs
Sustainability	Emission curve, fee-dominant revenue, no reward cliffs

Boing Network — Authentic. Decentralized. Optimal. Sustainable.